

Agricultural Crop-Yield Prediction

Machine Learning Capstone Project — Project Report

ML Task: Regression | **Target:** $\text{Yield} = \text{Production} \div \text{Area}$ | **Final Model:** LightGBM (Test $R^2 = 0.855$)

Dataset: District-wise, Season-wise Crop Production Statistics, Government of India Open Data Platform — 49,784 records, 1997–2014, 7 states, 112 districts, 80 crops.

Note on the model input rule: Production is never used as a model input, since it is the quantity used to derive the target itself ($\text{Yield} = \text{Production} \div \text{Area}$). Using it would leak the answer directly into the features.

1. Business Problem

Crop yield varies widely across India's districts, crops, and seasons, making it difficult for planners to know in advance which regions are likely to under- or over-perform in a given season. This project builds a regression model that predicts district-level crop yield from structural and historical information available before harvest.

Target Users / Stakeholders

- State and district agriculture departments planning subsidies and interventions
- Crop insurance providers assessing regional risk
- Agri-input companies (seed, fertilizer) planning distribution
- Farmer-facing advisory services

Expected Outcome & Business Value

A district × crop × season yield estimate that flags likely low-productivity regions early, enabling better-targeted subsidies, early-warning systems, and evidence-based crop planning — creating both business value (input suppliers, insurers) and social value (equitable resource allocation to under-performing districts).

2. Data Collection

Source: District-wise, Season-wise Crop Production Statistics, Government of India Open Data Platform (accessed via a static CSV export for this exercise).

- 49,784 raw records spanning 1997–2014
- 7 states, 112 districts, 80 distinct crops, 6 growing seasons
- Columns: State_Name, District_Name, Crop_Year, Season, Crop, Area, Production, Yield
- No personally identifiable or proprietary information is present in the dataset

3. Data Quality Assessment & Preprocessing

An initial audit found **zero missing values and zero duplicate rows** in the raw data, but 109 records with zero production and a small number of extreme outliers requiring treatment. The following steps were applied (see `src/preprocessing.py`):

- **Data-type correction:** year cast to integer; categorical fields cast to string and trimmed
- **Invalid-record removal:** rows with $\text{Area} \leq 0$ or $\text{Production} < 0$ dropped

- **Target recomputation:** $\text{Yield} = \text{Production} \div \text{Area}$, computed fresh rather than trusting the source 'Yield' column
- **Per-crop outlier treatment:** IQR-based capping ($\times 3$ IQR) computed separately for each crop, since yield scale varies hugely across crop types (e.g. sugarcane vs. pulses) — a single global threshold would have wrongly flagged entire high-yield crops as outliers
- **Leakage removal:** Production and the source Yield column dropped from the feature set entirely

Net effect: 49,784 $\rightarrow$ 49,170 rows (1.23% removed), all removals justified above.

4. Exploratory Data Analysis

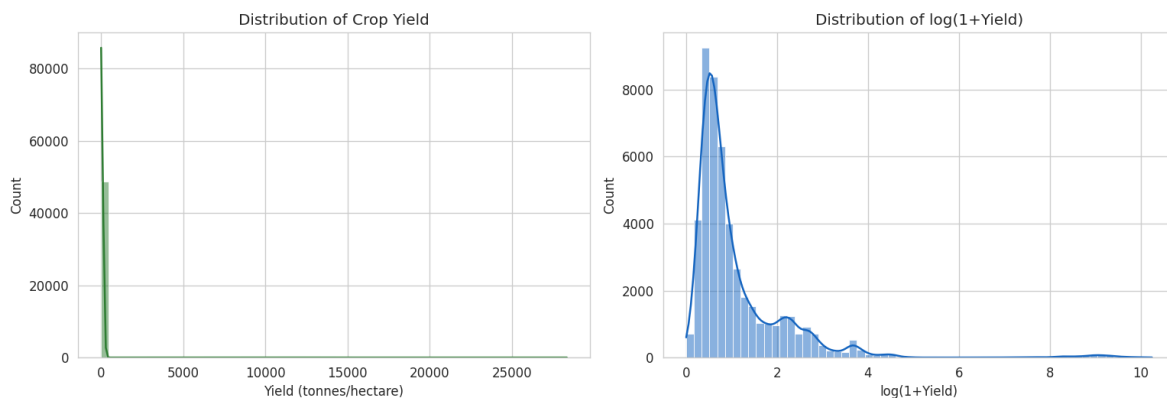


Figure 1. Yield is heavily right-skewed; a log transform (right panel) approximates normality, informing the choice of tree-based models that don't require normality assumptions.

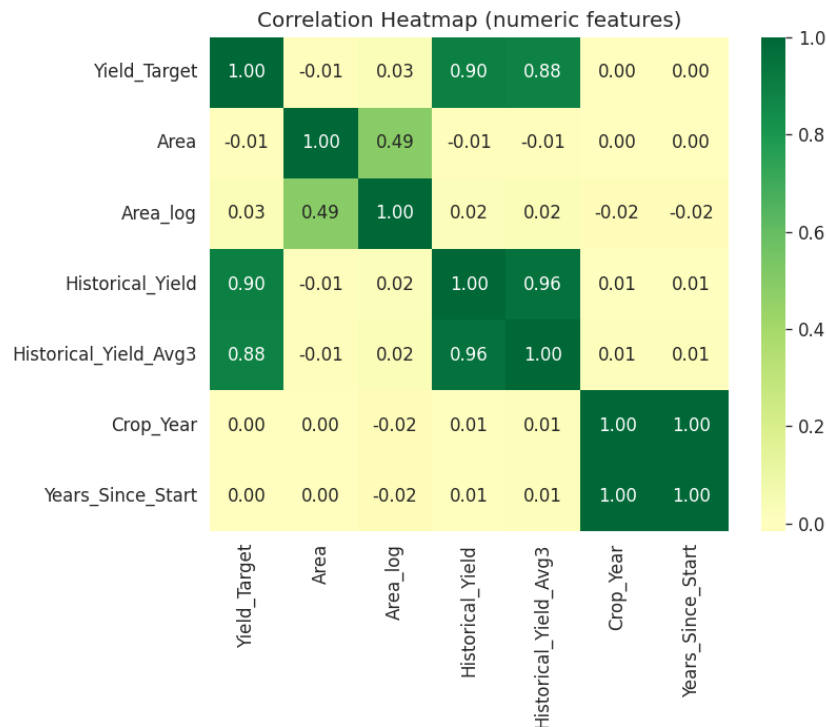


Figure 2. Correlation heatmap of numeric features. *Historical_Yield* and *Historical_Yield_Avg3* correlate strongly with the target ($r \approx 0.90$ and 0.88); raw cultivated Area has almost no linear relationship with yield.

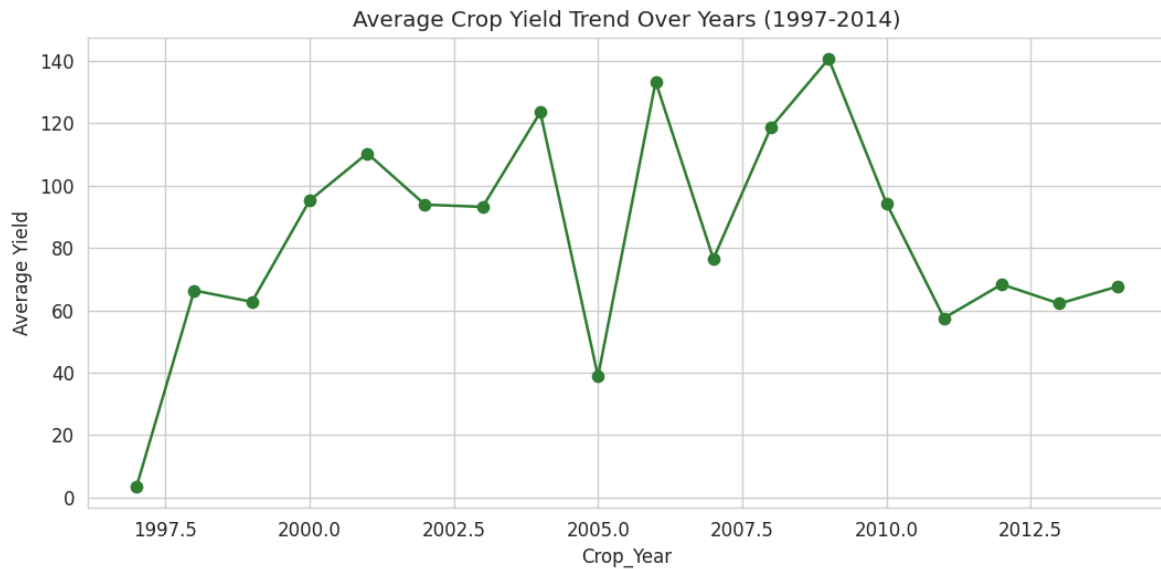


Figure 3. Average yield trend, 1997–2014 — relatively stable with year-to-year fluctuation, no strong long-term drift.

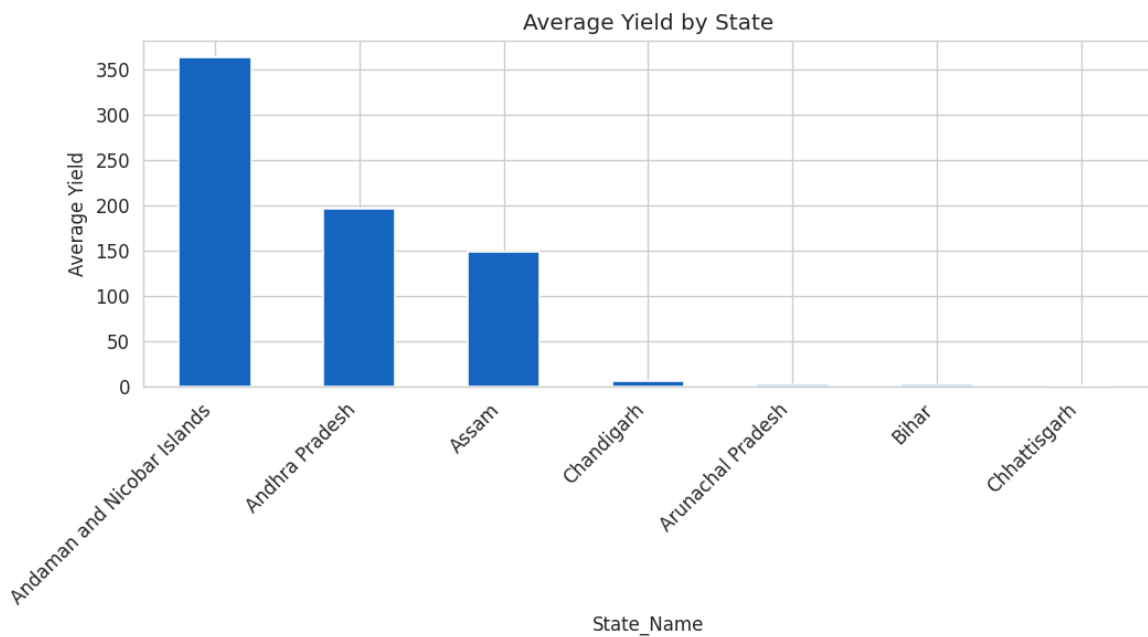


Figure 4. Average yield by state — Andaman & Nicobar Islands and Andhra Pradesh show the highest averages, partly influenced by high-yield-unit crops discussed below.

Key EDA finding: Coconut and the 'Whole Year' season show unusually high yield values because some crops in this dataset are recorded in count units (e.g. number of nuts) rather than tonnes. This is a labeling quirk of the source data worth flagging to stakeholders, not a data error to silently 'fix'.

5. Feature Engineering & Feature Selection

Historical performance was engineered as the primary predictive signal, computed strictly from *past* years to avoid leakage:

- **Historical_Yield** — lag-1 yield for the same State + District + Crop + Season combination
- **Historical_Yield_Avg3** — trailing 3-year average yield for the same combination
- **Area_log** — $\log(1+\text{Area})$, taming the heavy right-skew in cultivated area
- **Years_Since_Start** — a linear year-trend feature

Missing history (the first year a State-District-Crop-Season combination appears, ~11% of rows) is filled using **training-set-only** crop averages, computed after the train/test split — ensuring no future information leaks backward into earlier predictions.

Final feature set: State_Name, District_Name, Crop, Season (categorical, one-hot encoded) and Area_log, Historical_Yield, Historical_Yield_Avg3, Years_Since_Start (numeric, scaled). Production and the source Yield column are excluded as leakage.

6. Data Preparation & Validation Strategy

- **Time-based split:** train on years < 2012 (40,021 rows), test on 2012–2014 (9,149 rows) — chosen over a random split because Historical_Yield is a lag feature and the data has a temporal axis; a random split would let future years leak into earlier ones
- **Leakage-safe preprocessing:** scaling and one-hot encoding fit only on the training fold, inside an sklearn Pipeline + ColumnTransformer
- **3-fold cross-validation** on the training set for a stability check alongside the single held-out test score

7–9. Model Development, Tuning & Evaluation

Ten regression algorithms were trained and compared on identical train/test splits:

Model	Test R ²	Test RMSE	Test MAE	Test MAPE (%)
LightGBM	0.855	384.186	30.556	239.03
Random Forest	0.854	385.003	25.209	151.796
Gradient Boosting	0.834	409.984	29.557	298.631
XGBoost	0.799	451.709	41.309	567.001
Decision Tree	0.776	477.099	39.41	501.398
Ridge Regression	0.749	505.209	61.219	2280.812
Lasso Regression	0.748	506.224	60.58	2184.4

Model	Test R ²	Test RMSE	Test MAE	Test MAPE (%)
Linear Regression	0.747	506.441	61.549	2266.438
K-Nearest Neighbours	0.742	511.484	42.933	472.706
AdaBoost	0.498	714.055	237.708	19372.81

Genuine finding, not an error: AdaBoost produced a negative cross-validated R² (-3.47), meaning its default weak learners and loss weighting are poorly suited to this heavy-tailed, high-cardinality regression target. It was tested per the assignment's algorithm list but correctly not selected — models were chosen for suitability, not to force a particular result.

Hyperparameter Tuning

RandomizedSearchCV (3-fold CV, training data only) was run on the two strongest baseline models — LightGBM and Random Forest:

Model	Train R ²	Test R ²	Test MAE	Test RMSE	Overfit Gap
Random Forest (Tuned)	0.98	0.847	30.922	394.168	0.133
LightGBM (Tuned)	0.928	0.78	40.002	472.137	0.147

Neither tuned configuration beat the untuned baseline LightGBM (Test R² 0.855) on the held-out test set. The pipeline automatically keeps whichever model performs best, so the baseline LightGBM was retained as the final model — an honest outcome reported as-is rather than forcing an 'improvement' that wasn't real.

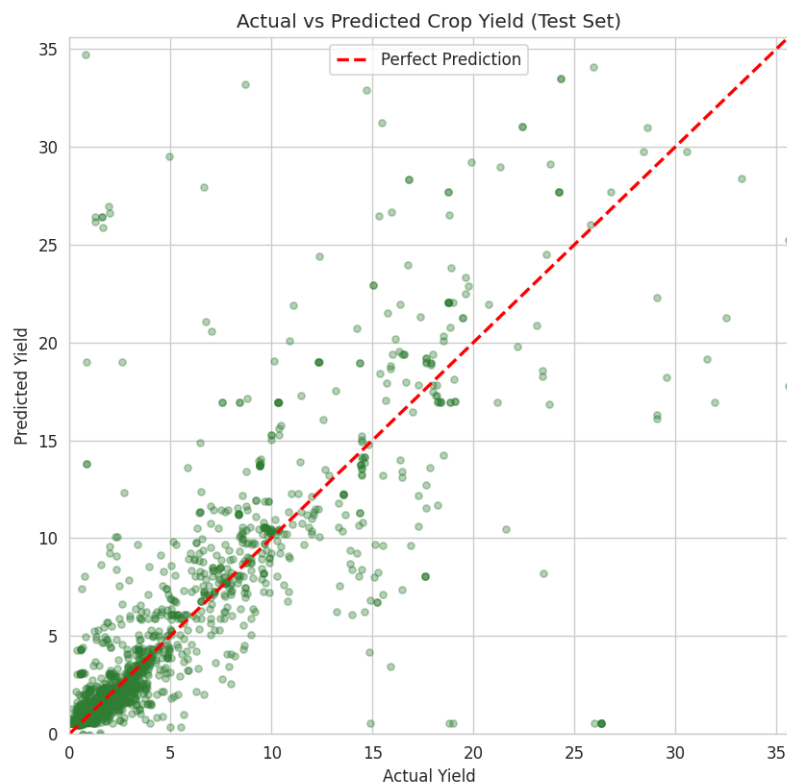


Figure 5. Actual vs predicted yield on the held-out 2012–2014 test set. Points cluster near the diagonal for the bulk of low/mid-yield records; scatter increases at high yield values.

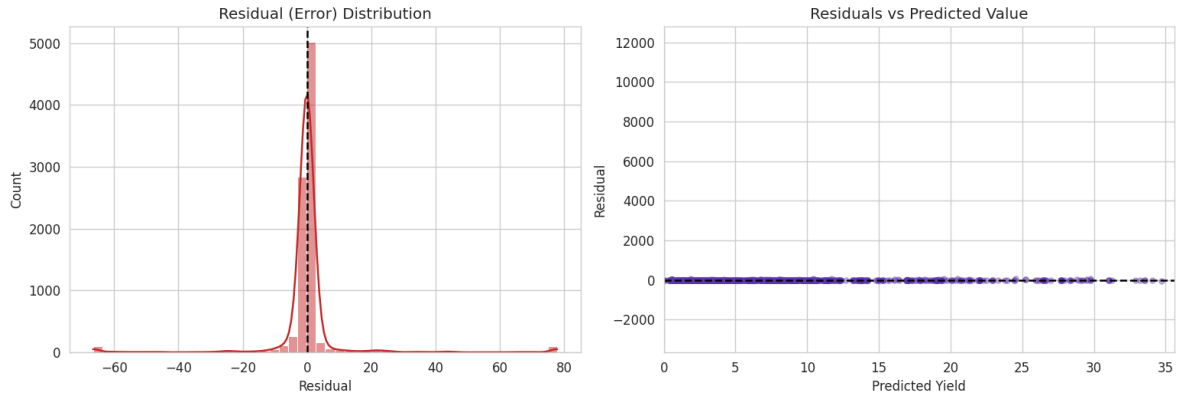


Figure 6. Residual distribution (left) is roughly centered at zero; residuals vs predicted (right) show mild heteroscedasticity at higher predicted yields.

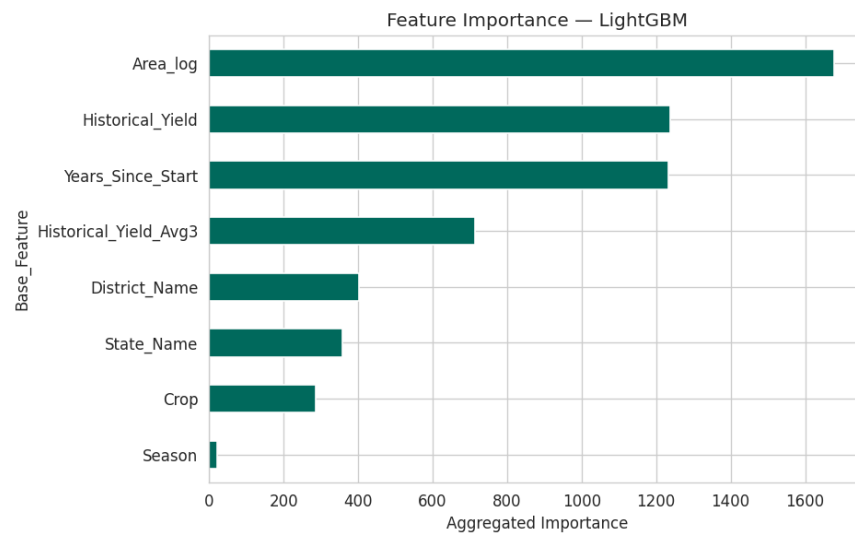


Figure 7. Aggregated feature importance for the final LightGBM model. Area_log, Historical_Yield, and Years_Since_Start dominate, confirming that past performance and field size are the strongest signals.

10. Final Model Summary

Metric	Value
Model	LightGBM
Train R ²	0.963
Test R ²	0.855
Test RMSE	384.19
Test MAE	30.56
Test MAPE	239.03%
Overfit Gap (Train–Test R ²)	0.108
Training Time	0.57 sec

11. Prediction & Result Interpretation

The final model was used to identify high- and low-productivity districts based on average actual yield in the test period (2012–2014):

Top 5 High-Productivity Districts

State	District	Avg. Actual Yield
Andhra Pradesh	SRIKAKULAM	466.4
Andhra Pradesh	KRISHNA	463.7
Andhra Pradesh	GUNTUR	431.6
Andhra Pradesh	SPSR NELLORE	403.1
Andhra Pradesh	WEST GODAVARI	379.8

Bottom 5 Low-Productivity Districts

State	District	Avg. Actual Yield
Chhattisgarh	BILASPUR	1.45
Bihar	LAKHISARAI	1.66
Chhattisgarh	BIJAPUR	1.74
Chhattisgarh	BALOD	1.75
Chhattisgarh	DANTEWADA	1.76

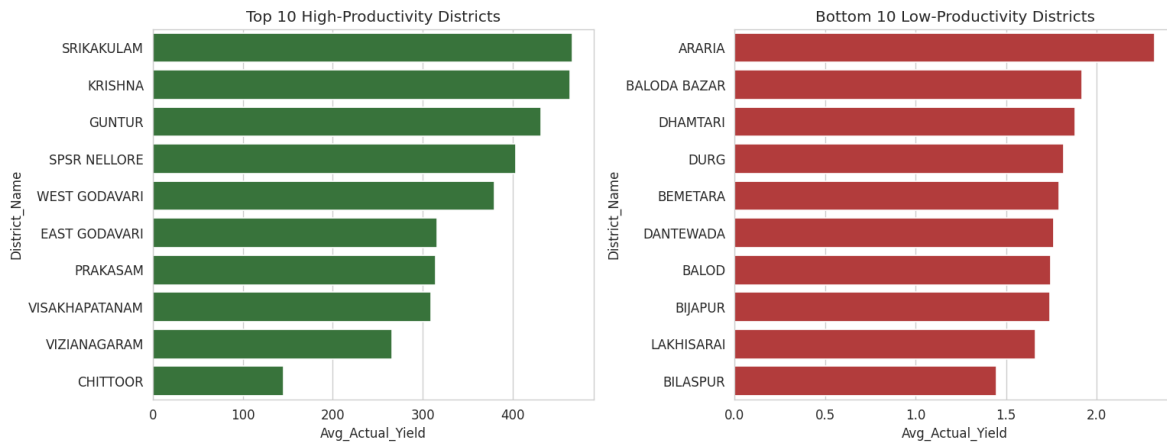


Figure 8. Top and bottom 10 districts by average actual yield (test period).

Limitations: the model relies heavily on historical yield for the same combination; genuinely new combinations fall back to a crop-level average and are less reliable. Crops recorded in non-tonne units (e.g. coconut) inflate averages for a few districts and should be modeled separately in a production system.

Future improvements: incorporate rainfall/weather and soil-quality data, model unit-mismatched crops separately, and add explicit prediction intervals via quantile regression.

12. Responsible AI & Ethical Considerations

- **Sensitive data:** none — only aggregated, public agricultural statistics are used
- **Bias check:** mean absolute error varies by state (see reports/error_by_state.csv), largely driven by how many historical records exist per state; this is disclosed rather than hidden
- **Fairness:** predictions are a planning aid, not a certified yield guarantee, and should not be used to unilaterally deny support to any district
- **Guardrails:** the dashboard explicitly labels all predictions as estimates

13. Reproducibility & Code Quality

- Modular scripts in src/, one responsibility per file, orchestrated by src/run_pipeline.py
- Random seeds fixed (random_state=42) throughout for reproducibility
- No hard-coded credentials; the project uses no external API keys
- requirements.txt provided for exact environment reproduction
- A fully executed Jupyter notebook (notebooks/Crop_Yield_Capstone.ipynb) and an interactive Streamlit dashboard (dashboard/app.py) accompany this report

14. Conclusion

The final model — **LightGBM**, achieving Test $R^2 = 0.855$ and Test RMSE = 384.2 on a chronologically held-out 2012–2014 test set — confirms that district-crop-season historical yield is the strongest available signal for near-term yield forecasting in this dataset. The full pipeline, from raw data to an interactive dashboard, is reproducible end-to-end via the provided scripts and notebook.

Dataset: Government of India Open Data Platform — District-wise, Season-wise Crop Production Statistics. Built with pandas, scikit-learn, XGBoost, LightGBM, Plotly, and Streamlit.